

TERA Memo Architecture

Documentation Status	Final - updated for 8/18/25 submission
Review Date	4/4/25
BIP Tenant Boundary	Compensation and Pension UI
BIP Tenant Component	tera
System Acronym	CPUI
VASI ID and Link	3028
BIP Tenant App(s)	bip-cpui-tera-memo-ui
GitHub Link(s)	https://github.com/department-of-veterans-affairs/bip-tera-memo-automation https://github.com/department-of-veterans-affairs/bip-cpui-tera-memo-ui
BIH Link(s)	Compensation and Pension UI
MSR Links(s)	2429
Solution and support Links	Solutioning: TERA Memo Solution Document
Sponsoring organization	VACO
Number of users	Approx. 30k potential unique users; Through-put of total users on any given day approximately 5k.
Estimated Monthly Cost	Estimated monthly costs: • AWS SageMaker: ~\$7,000 • CPUI Namespace: TBD
Privacy	PII: Veterans name and file number; non-veterans are not applicable to TERA Memo PHI: Medical data from documents in eFolder is reviewed but the data itself is not stored, only references to documents. PHI may be stored based on the final text submissions made by the user.
Will need 508 Compliance	Yes
Deployment date	26 January 2025
eMASS ID (System Name)	2049 (BIP Compensation and Pension User Interface)

Introduction

The TERA Memorandum form enables the Department of Veteran Affairs to expedite TERA Memorandum processing, incurring significant time savings for Veteran Service Representatives (VSR) reviewing as well as the Veteran claims processing. Additionally, TERA Memorandum Automation serves to solution and implement technology focused on Artificial Intelligence initiatives to provide the Department of Veteran Affairs with software that can offer problem-solving and decision recommendation capabilities in support of streamlining and standardizing the TERA Memorandum claim review process.

Key Definitions

- **TERA:** Toxic Exposure Risk Activity

- **VSR:** Veteran Service Representative who supports claim processing of TERA Memorandums
- **AWS SageMaker:** Allows for building and deploying models
- **Artificial Intelligence (AI) systems** are defined as: "Any artificial system that performs tasks under varying and unpredictable circumstances without significant human oversight, or that can learn from experience and improve performance when exposed to data sets." AI Systems Generally function by generating predictions, classifications or recommendations, or generating new content from inputs provided by the user or underlying applications (VA AI Inventory).
- **Optical Character Recognition (OCR):** technology that converts scanned documents and images into machine-readable text. VBMS SmartSearch solution provides this capability.

Key Capabilities

- Enhance efficiency of completing TERA Memorandum forms with available data from a Veteran's Claim Evidence.
- Provide pathway for future AI use cases on BIP Platform.

Intended Value

- The TERA Memorandum Automation functionality intends to deliver the value of enhanced efficiency to VSRs processing TERA Memorandum forms with the potential to 675,000 annual labor hours for users to spend on higher value activities that require their expert judgement.

Key Features

The below key features collectively support the MVP Scope delivery of TERA Automation ART as outlined in

 **BPT-4018** - Capability to Automatically Populate TERA Memo Questions Using Claims Evidence Data DONE .

- Modernized TERA Memorandum form User Interface with 508 Compliance
- Auto-Population button at the top of the TERA Memorandum UI to auto-populate certain questions within the form using evidence stored in the Veteran's Claim Evidence
- Add links to specific locations within Claim Evidence documents for manual review for each auto-populated answer
- Add confidence score for each auto-populated answer
- Gathering of metrics for submitted TERA Memorandum answer selections
- Monitoring Dashboard for tracking performance
- Feature flag functionality to enable/disable: Legacy & Modernized forms, Auto-Population Button, & Metrics gathering

AI Inventory Alignment

TERA Memorandum automation aligns to the VA AI Inventory use case of Human-Like Capabilities and is Rights Impacting. The TERA Memorandum Automation VA AI Inventory use case was submitted in August 2024.

- **Human-Like Capabilities:** Developed in software, hardware or other context to solve tasks requiring human-like perception, cognition.
- **Rights Impacting:** AI Systems whose outputs form the basis for decisions or actions that have legally significant effects on an individual's or entity's civil rights, liberties, privacy, or access to essential opportunities.

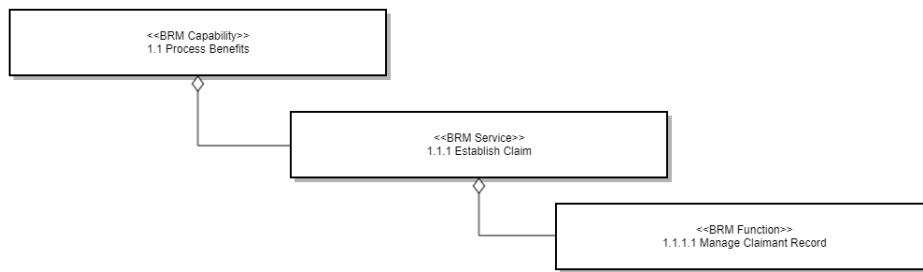
Use Case View

As a Veteran Service Representative (VSR), I need to process Veteran claims for Toxic Exposure Risk Activity (TERA) so that I can identify if a Veteran meets the criteria for TERA related activities.

Capability Viewpoint Vision (CV-1) (VASI Description)

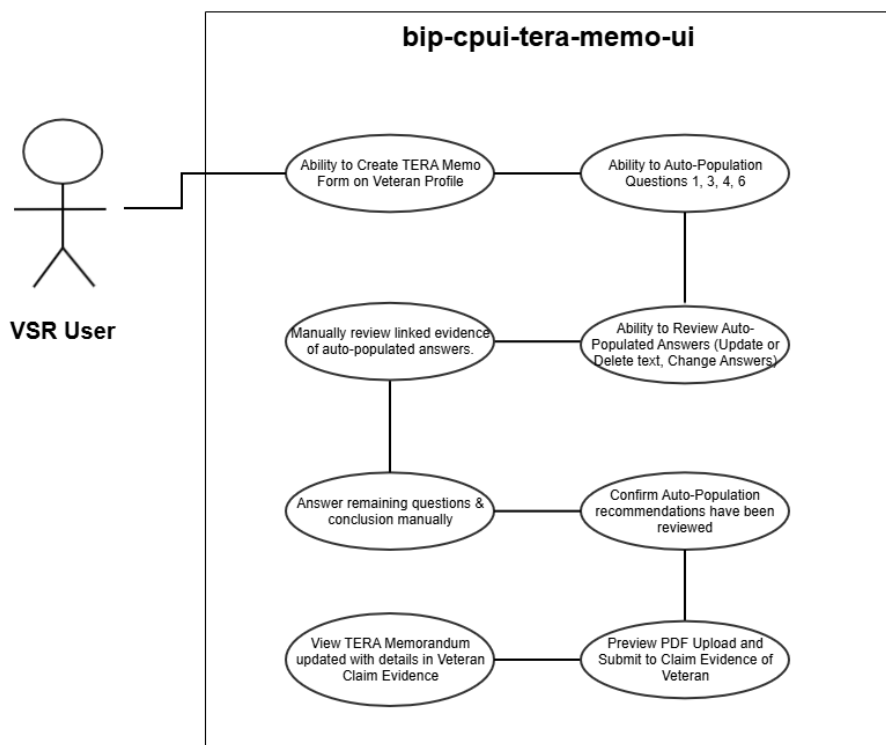
TERA Memorandum Automation (TERA Automation) resides on Benefits Integrated Platform as a Service (BIP- #2295) and is under the Compensation and Pension UI (CPUI-3028) Product. TERA Memo UI enables VSR to leverage Artificial Intelligence in completing initial TERA Memo processing. VSRs will be able to initiate auto-population of evidence gathered that is relevant to TERA Memo intake. Given a confidence score as well as links to specific locations within document evidence for manual review, VSRs can efficiently assess and certify the validity of evidence gathered.

Capability Viewpoint Taxonomy (CV-2)



Use Cases

The below diagram outlines the expected behavior of the Veteran Service Representative (VSR) while within VBMS completing a TERA Memorandum form.



User Interface Functionality

The TERA Memorandum Form was modernized to follow updated VBMS standards for user interface and experience in line with 508 compliance standards. Design updates include:

- Modernized User Interface Design
 - Veteran Information pre-populated
- Addition of auto-population user interface components including:
 - Auto-Population button
 - Auto-Population information modal on buttons election
 - Checkbox of manual review of auto-populated recommendations
- Policy change updates to TERA Memorandum for questions and answers.
- Access to the TERA Memorandum Form is controlled through VBMS Document Policy on the Veteran Profile that will show the TERA Memo as a drop-down action when the end user has the ability to Upload a document.

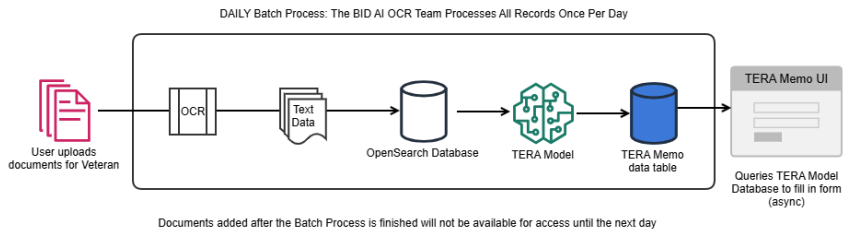
Auto-Population Functionality

TERA Automation utilizes algorithms to provide recommended decisions to support completing the TERA Memorandum form.

- Input: Data & eligibility sources
 - Data extracted from Veteran's Claim Evidence documents via C&P Smart Search Optical Character Recognition (OCR) scans
 - Data extracted from Veteran's ILER IES record
 - PACT Act Standard Operation Procedures
 - VA Guidance from the M21-1 Adjudication Procedures Manual
- Model:
 - Algorithms that analyze and interpret the input data, leveraging AWS SageMaker via the ML-MATS Platform, applying the eligibility requirements and guidelines from the PACT Act SOP and M21-1 to the Veteran's Claim Evidence
- Output: Evidence in support of "Yes" answer to Questions 1, 3, 4 and/or 6:
 - Determination of "Yes" selection on form
 - Open-Text field response with supporting details
 - Linked to specific locations within document evidence identified document(s) and Confidence Score for manual review

Model Processing

- Daily Batch Process
 - a. As part of this daily batch process, documents added to the evidence folder are converted from Image or PDF to text files. These text files are then added to the OpenSearch database and indexed.
 - b. The final step in this daily process is to run the TERA Model, which queries evidence in the OpenSearch Database and then writes the answers to the TERA Memo data table.
 - c. This data table can then be queried by the TERA Memo UI to support auto-population.



Main Actors

The following table describes the actors listed in the diagram as a user in the scope of the TERA Memorandum Automation architecture.

Name	Description
VSR	The individual tasked with VBMS claim processing including the TERA Memorandum form.

Systems Details

The following table describes the system details for TERA Memorandum Automation.

System	Description
VBMS CPUI	Contains the React / Micro Front End TERA Memo UI
Claim Evidence (Claim Evidence API)	Source of TERA evidence including a SmartSearch endpoint that houses OCR results from eFolder documents
VBMS Core DB	Repository of Veteran data. Leveraged during the auto-population process
ML-MATS Platform (AWS SageMaker)	Houses AI models used to automate evidence gathering and uses CICD pipelines to deploy to upper environments

Main User Functions

VSR will be able to use the auto populate feature to automate evidence gathering in support of TERA Memo completion.

Actor	Description	Usage
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VSR	The VSR is tasked with completing the TERA Memorandum form questions in VBMS.	After selecting the auto-population button, the VSR will manually review the recommended answers and linked evidence documents retrieved from artificial intelligence (AI) models along with confidence score of results and update the answers as needed before previewing and submitting the form.
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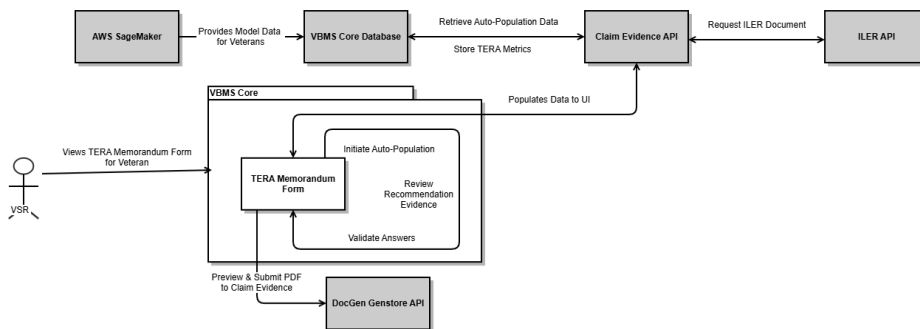
TERA Memorandum Questions	Details
Question 1	Question 1 text: "Does the Veteran qualify for a presumption of exposure(s) for one or more of the following hazards during military service? (If Yes, provide details below.)" User Response Options: - "Yes, the Veteran was exposed to the following" - "No, the Veteran was not exposed to the following" If "Yes" is selected: The user must indicate at least one of the following hazards, presented with checkboxes (multiple selections are allowed): - Herbicide Agent - 2,3,7,8 - Tetrachlorodibenzodioxin (TCDD) - Radiation – Ionizing - Mustard Gas - Nitrogen, Sulfur Mustard, or Lewisite - Camp Lejeune Water Contamination - Perchloroethylene (PCE), trichloroethylene (TCE), vinyl chloride, benzene Under question 1 checkboxes, the user has the ability to enter a free text response, the following prompt is provided: "Evidence of exposure (include the subject and date of receipt of the cited documents):"
Question 2	Question 2A text: "Is there an Individual Longitudinal Exposure Record (ILER) entry for the Veteran? (If No, ensure documentation is uploaded in VBMS. If Yes, answer 2B and ensure that the ILER (Individual Exposure Summary (IES)) Report is uploaded into VBMS. Note: Name only, contractor, and civilian entries are not TERAs; select No in these circumstances.)" Question 2A user response options: - "Yes" - "No" Question 2B text: "Is there an ILER entry for an activity carried out by the Veteran while on active duty involving toxic exposure to occupational or environmental hazards? (If Yes, answer 2C as well. If No, select the type(s) of ILER entry(ies) of record that were determined to not involve toxic exposure during active duty, and provide justification for this determination in the free text box below. See "TERA Exception Job Aid" for more information on the circumstance-based exceptions.)" Question 2B user response options: - "Yes" - "No, the ILER entry(ies) do not involve toxic exposure (select all that apply)" Under the "No, the ILER entry(ies) do not involve toxic exposure (select all that apply)" response, there are two checkboxes: - "Do/does not corroborate toxic exposure to occupational or environmental hazards" - "Consist(s) of self-reported information that cannot be substantiated" Under question 2B checkboxes, the user has the ability to enter a free text response, the following prompt is provided: "Explanation of determination:" Question 2C text: "Is there any evidence of exposure over permissible limits or of concern (identified in red text) in the ILER report? (If Yes, ensure that pertinent evidence is bookmarked or annotated in VBMS. For more information, see Exposure Levels that Exceeded Permissible Levels in "ILER Guidance.")" "TERA Exception Job Aid" in the final sentence of the question contains a link to - TERA Exception Job Aid.pdf Question 2C user response options: - "Yes" - "No"

Question 3	Question 3A text: "Does the Veteran's VBMS eFolder contain the Persian Gulf War Service - Sec. 1117 flash and/or 1117 memo? (If No, answer 3B as well.)" Question 3A user response options: - "Yes" - "No" Question 3B text: "Is there evidence of record verifying that the Veteran served in a 38 C.F.R. § 3.317(e)(2) and/or Sec. 1117 location? (If Yes, provide details below.)" Question 3B user response options: - "Yes" - "No" Under question 3B, the user has the ability to enter a free text response, the following prompt is provided: "Evidence of service in a 38 C.F.R. § 3.317(e)(2) and/or Sec. 1117 location (include the subject and date of receipt of the cited documents):"
Question 4	Question 4A text: "Does the Veteran's VBMS eFolder contain the <i>Toxic Exposure – Sec. 1119 Covered Veteran</i> flash and/or 1119 memo? (If No, answer 4B. If Yes, answer 4C as well.)" Question 4A user response options: - "Yes" - "No" Question 4B text: "Is there evidence of record verifying that the Veteran served in a 38 C.F.R. § 3.320 and/or Sec. 1119 location? (If Yes, answer 4C and provide details below.)" Question 4B user response options: - "Yes" - "No" Question 4C text: "Does the Veteran's VBMS eFolder contain the <i>Toxic Exposure-Uzbekistan Deployed Veteran</i> flash? (If No, answer 4D.)" Question 4C user response options: - "Yes" - "No" Question 4D text: "Is there evidence of service at Karshi-Khanabad (K2)? (If Yes, provide details below.)" Question 4D user response options: - "Yes" - "No" Under question 4D, the user has the ability to enter a free text response, the following prompt is provided: "Evidence of service in a 38 C.F.R. § 3.320 and/or Sec. 1119 location or K2 (include the subject and date of receipt of the cited documents):"
Question 5	Question 5A text: "Is there evidence of other deployment related exposure in the eFolder (not already depicted in the questions above) which is consistent with the circumstances of the Veteran's service? (If Yes, provide details below and answer 5B as well.)" Question 5A user response options: - "Yes" - "No" Under question 5A, the user has the ability to enter a free text response, the following prompt is provided: "Evidence and details about deployment related exposure (include the subject and date of receipt of the cited documents):" Question 5B text: "Where in the eFolder does the deployment information exist (e.g., VIS, military personnel records, DD214, service treatment records, post-deployment health assessments, PIES responses, etc.)? (Ensure that the pertinent evidence is bookmarked or annotated in VBMS.)" Under question 5B, the user has the ability to enter a free text response.

Question 6	Question 6 text: "Is there evidence of non-deployment related exposure in the eFolder which is consistent with the circumstances of the Veteran's service? (If Yes, provide details below.)" Question 6 user response options: - "Yes" - "No" Under question 6, the user has the ability to enter a free text response, the following prompt is provided: "Evidence and details about non-deployment related exposure (must include a description of the specific toxic exposure(s) being conceded, to include any relevant location, dates, and length of each exposure, etc., and the subject and date of receipt of the cited documents):"
Conclusion	Conclusion part 1 text: "Did the Veteran participate in a TERA during active military service? (If Yes was answered to any of the above questions, then "Yes, the Veteran participated in a TERA" should be selected, and Part 2 should be answered as well.)" Conclusion part 1 user response options: - "Yes, the Veteran participated in TERA" - "No, the Veteran did not participate in a TERA" Conclusion part 2 text: "Is the TERA participation of record sufficient to trigger an examination/opinion under 38 U.S.C. § 1168(a)? (Important: If 2A is the only question on the TERA memorandum with a Yes response, then an examination and opinion under 38 U.S.C. § 1168(a) is/are not warranted and the No answer option should be selected below.)" Conclusion part 2 user response options: - "Yes" - "No, the only participation in a TERA that is established is based on an entry in an exposure tracking record system, such as ILER, that is deemed an exception under 38 U.S.C. § 1168(b)." - "No, the Veteran served in a location where exposure to herbicides is conceded. However, herbicide exposure is the only TERA and the condition has been determined by the Secretary to have no positive association with herbicide exposure." Under conclusion part 2, the user has the ability to enter a free text response, the following prompt is provided: "Identify the contention(s) in the text box below:"

Primary Systems High Level Operational Viewpoint (OV-1)

The OV-1 is a high-level graphical/textual description of the operational concept (what happens, who does what, in what order, to accomplish what goal). The purpose of OV-1 is to provide a quick, high-level description of what the architecture is supposed to do, and how it is supposed to do it (players /operations involved).



The primary components of the TERA Memo architecture are a React Front End, API, and AI components.

VBMS Resource Locations
White Rectangle - TERA application resource to resource being documented
Gray Rectangle - Existing application resource to resource being documented

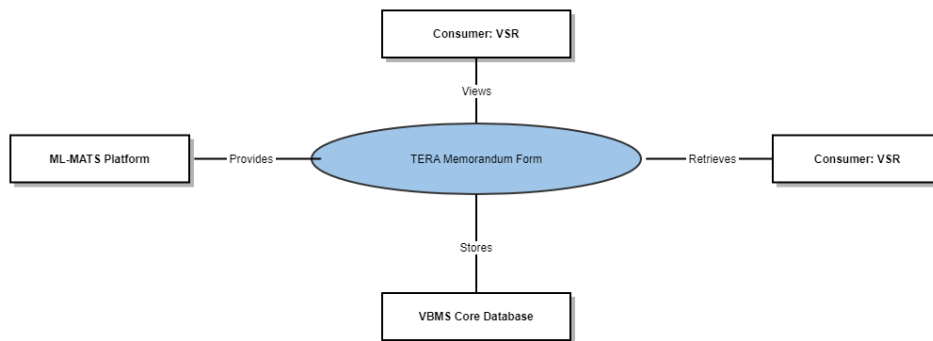
The table below describes the lines and arrows in the diagram above:

System A and VASI ID	System B and VASI ID	Interaction Description
AWS SageMaker	OpenSearch Endpoint	Enables the ability to run the TERA model against OpenSearch Veteran Dataset and store in VBMS Core Database

Claims Evidence API	VBMS Core Database	Pulls relevant data on Veteran from VBMS Core Database to support in population TERA Memo UI
Claims Evidence API	TERA Memo UI	Auto-population populates TERA Memo with relevant recommendation for Veteran
ILER API	TERA Memo UI	Retrieves Veteran ILER document used to provide data for auto-population
TERA Memo UI	CE API / CE UI	Ability for user to click evidence link for auto-populated recommendation to manually review specific locations in document
TERA Memo UI	DocGen GenStore API	Upload of completed TERA Memorandum Document to Claim Evidence
VBMS	TERA Memo UI	Initiate auto population Review Recommendation Validate answers

Operational Resource Flow (OV-2)

The OV-2 is a behavioral diagram that describes the Resource Flows exchanged between "Operational Performers." In other words, it shows the collaboration of main systems on the key data elements (e.g., system A sends a package while system B receives the package).



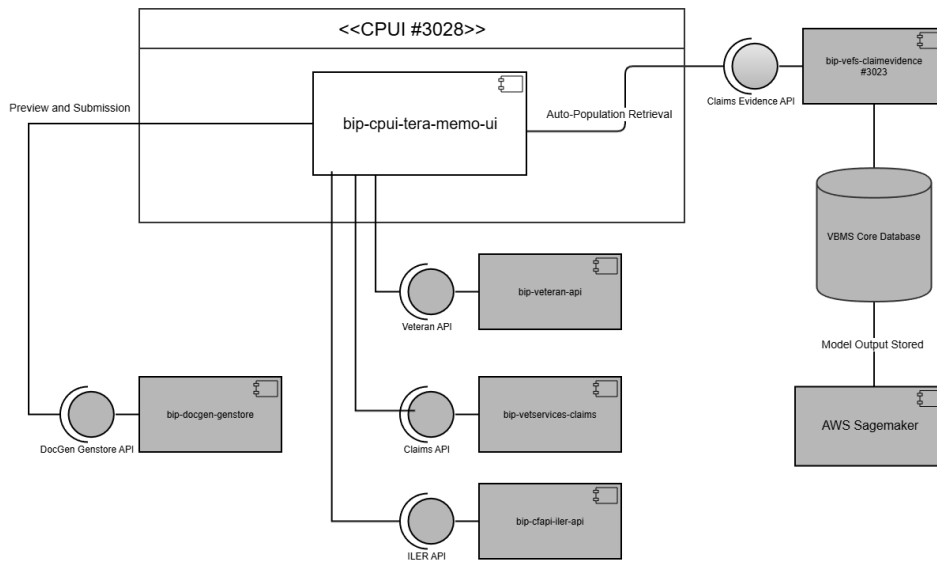
Logical View

Implementation of the modernized TERA Memo in VBMS involves three core concepts: User Interface, TERA Recommendation Model, & APIs.

- The TERA Memo UI is a React Micro FrontEnd (MFE) that communicates with the TERA Memo API and the existing Claim Evidence API.
- TERA Recommendation model runs nightly to abstract relevant information and input recommendations in the VBMS Core database for each question on the TERA Memorandum Form based on available data in a Veteran's Claim Evidence.
- The respective APIs then integrate with an Oracle DB for data retrieval. The database is populated by AI model outputs from AWS SageMaker with use of AWS Comprehend.

Systems Interface Description (SV-1)

The SV-1 provides information about the composition and interaction of Systems. Each system involved in the architecture should be represented, showing the VASI ID in the << >> Stereotype. Interfaces between systems must be displayed as lollipop notation, where the service provider has the API /Service shown as the lollipop and service consumers "eating" the lollipop. Dependencies or other relationships can also be shown as arrows as appropriate.



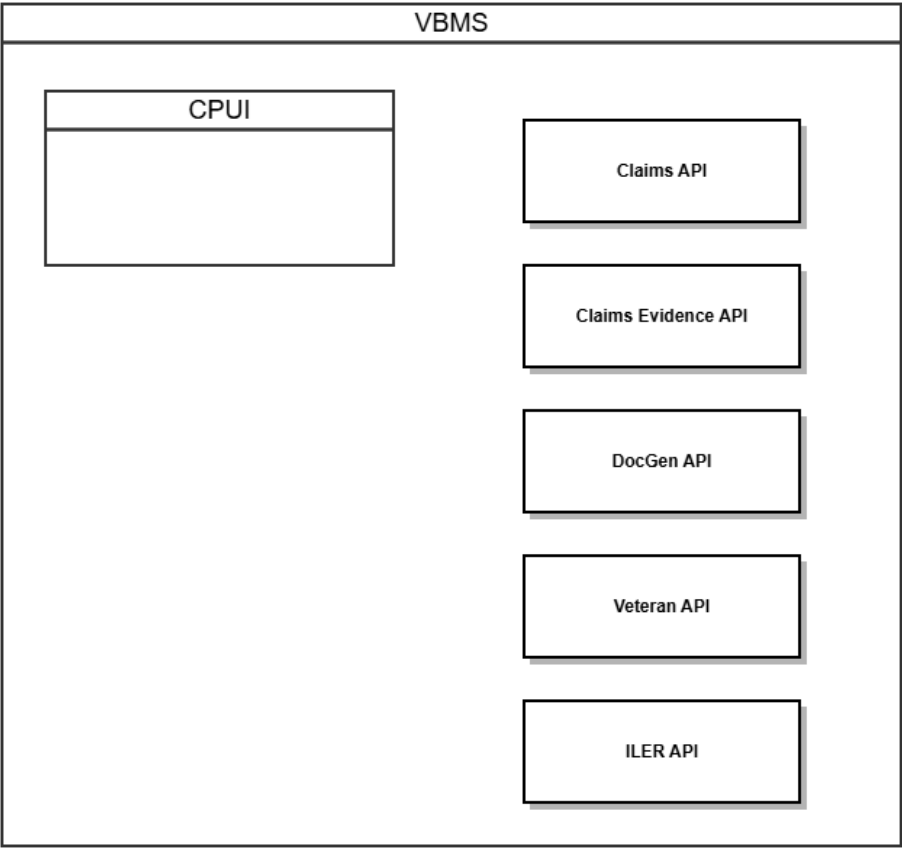
Gray Rectangle = External VBMS resource to resource being documented
 White Rectangle = Internal VBMS resource to resource being documented

Component References Table

Name	Description	Ports	Protocols
CPUI #3028	Houses bip-cpui-tera-memo-ui TERA Memorandum Form	443	https
AWS SageMaker	Builds and deploys models that store auto-population data output into VBMS Core Database	443	https
ILER API #3004	The ILER API is a service used to automate the upload process of ILER IES records into the VBMS eFolder. This functionality will allow the seamless integration and transfer of the Service member/Veteran's ILER IES record into VBMS. By replacing the current manual process, this service will minimize the opportunity for human error as well as the possibility of duplicate efforts.	443	https
VBMS #1728	VBMS Core Database (Model Output from AWS SageMaker via ML-MATS & Metric of Submitted data)	443	https
DocGen Genstore API #3000	Support creation & submission of PDF documents	443	https
BIA Claims API #3237	Claims API retrieves Veteran's Claim data to support ability to associate the TERA Memorandum to appropriate open claims during the Submit TERA Memorandum process.	443	https
CFAPI Veteran API #3004	Provides the current Veteran's Name and File number to populate the TERA Memorandum Form Veteran Details.	443	https
Claim Evidence API #3023	On Preview, DocGen API generates a PDF document including the generation of a Document ID and then DocGen API has a follow-on request to show the generated document the with equivalent fields representing the completed digital TERA Memorandum form in a new browser tab. On Submit, DocGen API generates a PDF and uploads to the Veteran's Claim Evidence (eFolder).	443	https

Services Context (SvcV-1) (VASI APIs)

The Svc-V1 Service Context's purpose is to describe and identify service composition and interaction. It shows a given Application Component, its services (SOAP, REST, RPC, EJB, Messaging), and its immediate service dependencies. The context for this diagram shows several elements:



Service /API	Operation ID	Description
Claims Evidence (CE) API		On Auto-Population, CE API retrieves information from VBMS database table of TERA recommendations (OCR_EXTRACT_DATA) to auto-populate the TERA Memorandum form and retrieve relevant documents for including in links to specific evidence locations supporting manual review. ▪ Leveraging CE API with annotations/custom viewer to link directly to specific text within a document. On Submission of TERA Memorandum form, CE API will write metrics of submitted answers to database table (VBMSUI. TERAMEMOUPLOADMETRICS).
Claims API		Claims API retrieves Veteran's Claim data to support ability to associate the TERA Memorandum to appropriate open claims during the Submit TERA Memorandum process.
DocGen Genstore API		On Preview, DocGen API generates a PDF document including the generation of a Document ID and then DocGen API has a follow-on request to show the generated document the with equivalent fields representing the completed digital TERA Memorandum form in a new browser tab. On Submit, DocGen API generates a PDF and uploads to the Veteran's Claim Evidence (eFolder).
Veteran API		Called to get the current Veteran's Name and File number to populate the TERA Memorandum Form Veteran Details.
ILER API		Called to retrieve Veteran's ILER IES record to support auto-population of question 2.

Services Resource Flow (SvcV-2)

A SvcV-2 specifies the Resource Flows between Services and may also list the protocol stacks used in connections. Its purpose is to give a precise specification of a connection between Services, showing specific resource operation dependencies.

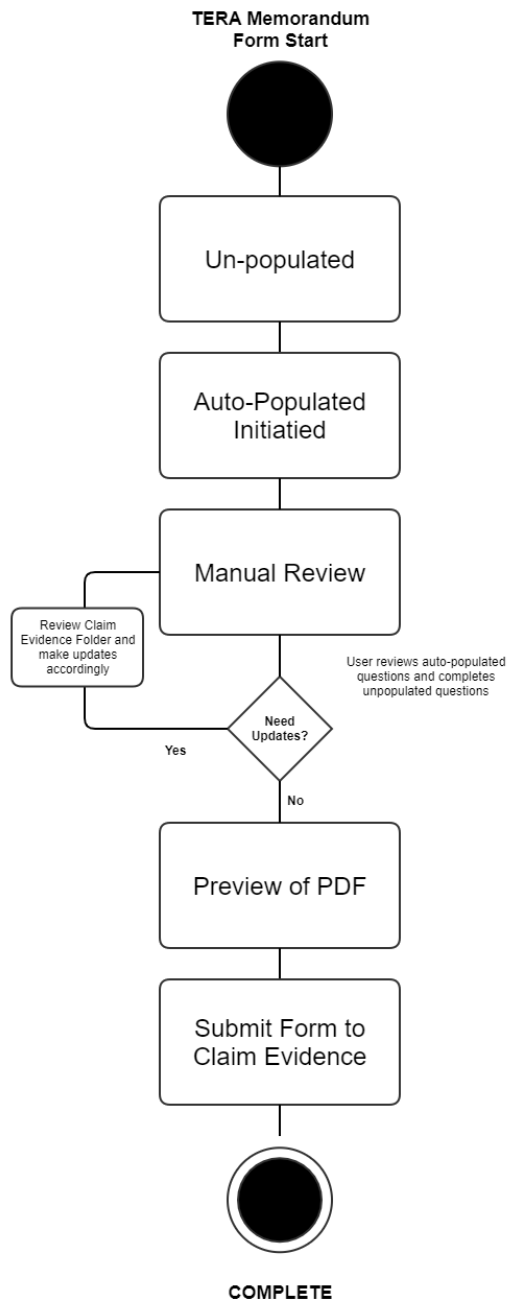
Source System	Source Service	Source Operation	Target System	Target Service	Target Operation
TERA Memo UI	Veteran API Resource	getVeteran	CFAPI	Veteran API Service	findVeteran
TERA Memo UI	ILER API Resource	requesttlerDocument			
TERA Memo UI	TeraMemoController	getTeraMemoData		Claimant Service	retrieveFlashes
TERA Memo UI	TeraMemoController	postTeraMemoMetrics			
TERA Memo UI	FileController	getData			
TERA Memo UI	Claims Resource	getClaimsByVeteran			
TERA Memo UI	GenstoreFeignClientOther	generateDocument			

Internal and External Systems-Systems Matrix (SV-3/SvcV-3) (VASI Interfaces)

System/Service A	Service / System B	Internal or External	Consumer or Dependency (in relation to System A)
Claim Evidence: CE API (3023)	CPUI: TERA Memo UI (3028)	Internal VBMS	Dependency
BIA Benefits Services: Claims API (3232)	CPUI: TERA Memo UI (3028)	Internal VA	Dependency
DocGen: Genstore API (3000)	CPUI: TERA Memo UI (3028)	Internal VBMS	Dependency
CFAPI: Veteran API (3004)	CPUI: TERA Memo UI (3028)	Internal VBMS	Dependency
CFAPI: ILER API (3004)	Claim Evidence: CE API (3023)	Internal VBMS	Consumer
CPUI: MFE Auth Util (3028)	CPUI: TERA Memo UI (3028)	Internal VBMS	Dependency

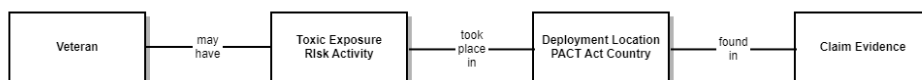
State Transitions (OV-6b)

The OV-6b's purpose is to indicate how the status of key objects change in response to activities in the system. It shows how resources change over time. Any object that has a discrete set of statuses or a lifecycle requires a state-transition diagram.

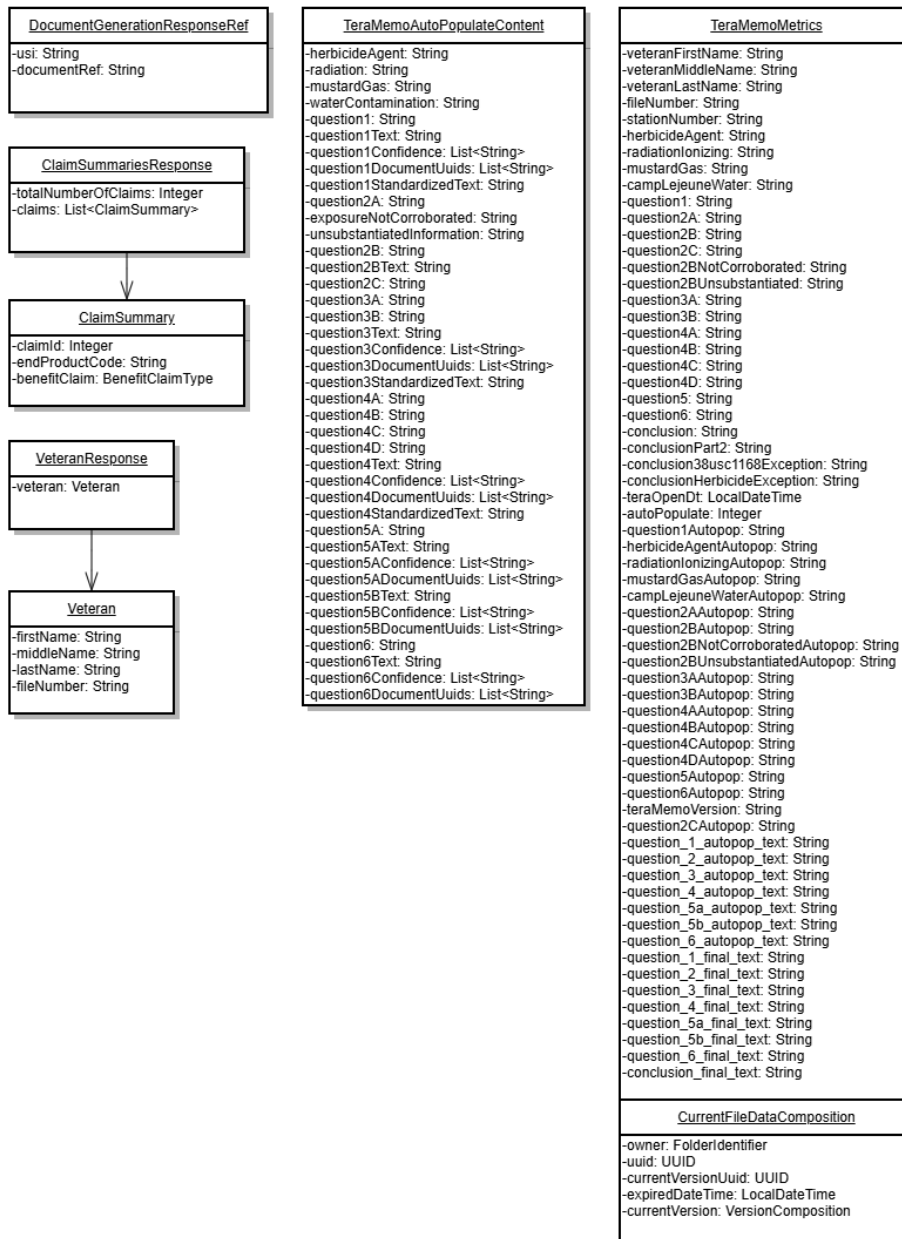


Data Models

Conceptual Data Viewpoint Model (DIV-1)



Logical Data Viewpoint Model (DIV-2)



Physical Data Viewpoint Model (DIV-3)

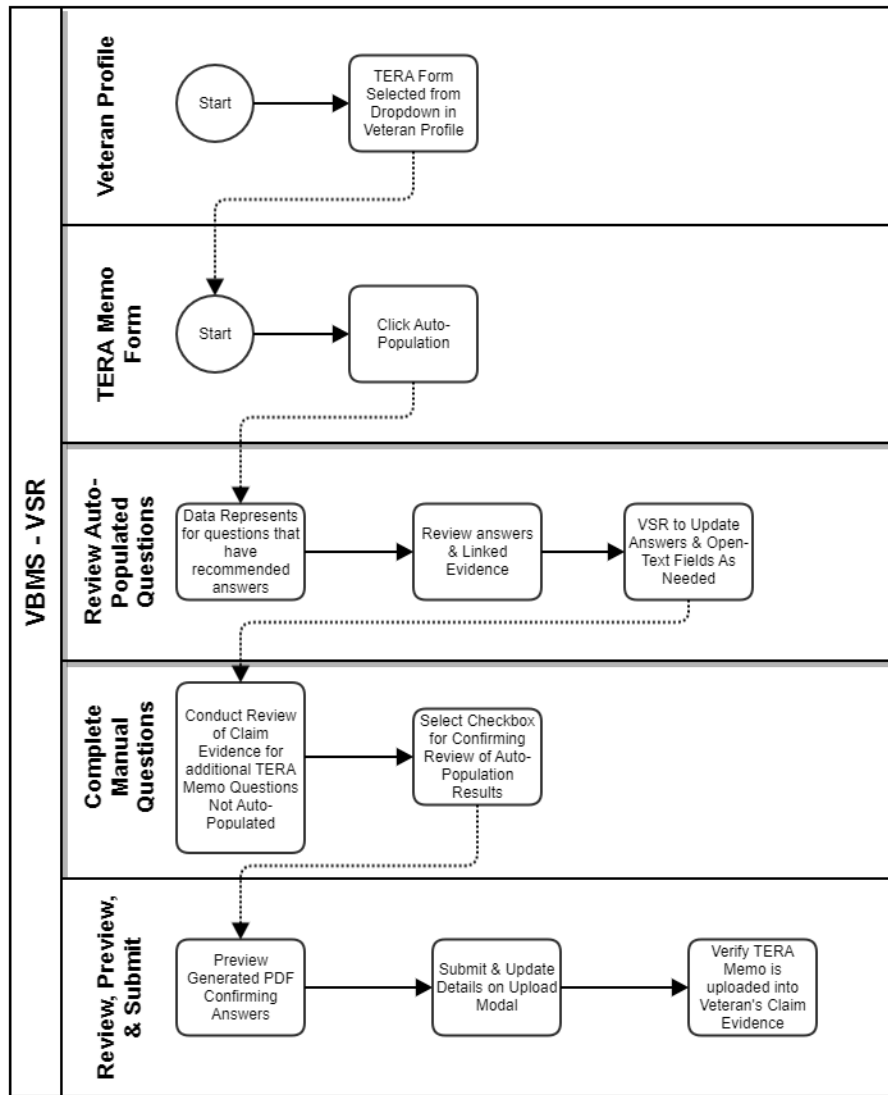
OCR TEXTTRACT DATA
ID (PK)
UPDATE_DATE
STATUS
DATA
PARTICIPANT_ID

TERA STANDARDIZE TEXT LU
ID (PK)
QUERY_ID
DOCUMENT_TYPE
UPDATE_DATE
TERA_STANDARDIZED_TEXT

TERAMEMOUPLOADMETRICS
ID (PK)
VETERAN_LAST_NAME
VETERAN_FIRST_NAME
VETERAN_MIDDLE_INITIAL
FILE_NUMBER
UPLOAD_DT
USER_ID
STATION_NUMBER
QUESTION_1
HERBICIDE_AGENT
RADIATION_IONIZING
MUSTARD_GAS
CAMP_LEJEUNE_WATER
QUESTION_2A
QUESTION_2B
QUESTION_3A
QUESTION_3B
QUESTION_4A
QUESTION_4B
QUESTION_5
QUESTION_6
CONCLUSION
QUESTION_2C
QUESTION_2B_NOT_CORROBORATED
QUESTION_2B_UNSUBSTANTIATED
CONCLUSION_PART2
CONCLUSION_USC1168_EXCEPTION
CONCLUSION_HERBICIDE_EXCEPTION
QUESTION_4C
QUESTION_4D
TERA_OPEN_DT
AUTO_POPULATE
QUESTION_1_AUTOPOP
HERBICIDE_AGENT_AUTOPOP
RADIATION_IONIZING_AUTOPOP
MUSTARD_GAS_AUTOPOP
CAMP_LEJEUNE_AUTOPOP
QUESTION_2A_AUTOPOP
QUESTION_2B_AUTOPOP
QUESTION_3A_AUTOPOP
QUESTION_3B_AUTOPOP
QUESTION_4A_AUTOPOP
QUESTION_4B_AUTOPOP
QUESTION_5_AUTOPOP
QUESTION_6_AUTOPOP
QUESTION_2B_NOT_CORROBORATED_AUTOPOP
QUESTION_2B_UNSUBSTANTIATED_AUTOPOP
QUESTION_4C_AUTOPOP
QUESTION_4D_AUTOPOP
TERA_MEMO_VERSION
QUESTION_2C_AUTOPOP
QUESTION_1_AUTOPOP_TEXT
QUESTION_2_AUTOPOP_TEXT
QUESTION_3_AUTOPOP_TEXT
QUESTION_4_AUTOPOP_TEXT
QUESTION_5A_AUTOPOP_TEXT
QUESTION_5B_AUTOPOP_TEXT
QUESTION_6_AUTOPOP_TEXT
QUESTION_1_FINAL_TEXT
QUESTION_2_FINAL_TEXT
QUESTION_3_FINAL_TEXT
QUESTION_4_FINAL_TEXT
QUESTION_5A_FINAL_TEXT
QUESTION_5B_FINAL_TEXT
QUESTION_6_FINAL_TEXT
CONCLUSION_FINAL_TEXT

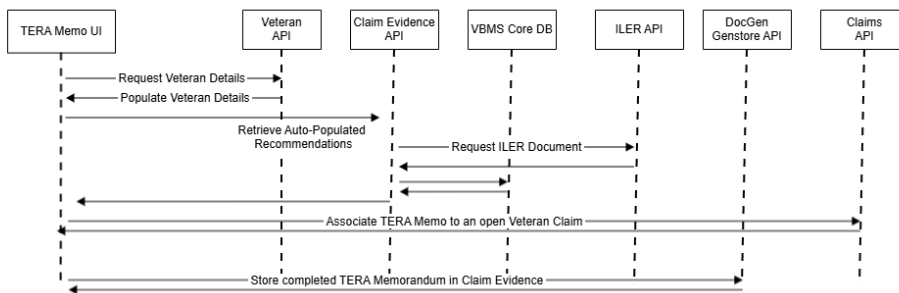
Process View (Business Process Model)

Main Business Process Model (OV-6d)



Process Sequence - Systems Event-Trace Description (SV-10c)

The below diagram outlines the interactions between functional resources.



Implementation View

Primary System Technologies (SV-9) (VASI Technology Components)

For standard tools see [VBMS SV-9 Master Table](#).

Software Type	Software Vendor	Software Name	Software Version	TRM Link	TRM ID
Development Tool	OpenJS Foundation	Node.js	Latest	Link	6716
Cloud Services/Server Virtualization	Amazon	Sagemaker	N/A	N/A	N/A
Cloud Services/Server Virtualization	Amazon	Comprehend	N/A	N/A	N/A
Development Tool	Python Software Foundation	Python	Latest	Link	5702

Deployment View

Environment Mapping

See [VBMS Environments Info Page](#) for a current list of environments and descriptions. TERA Memo UI is deployed to all VBMS environments.

System Configuration

Feature Flag	Purpose	Location (Github)
TERA Memorandum Form	Enables ability to toggle on / off the TERA Memorandum Form from VBMS Claim Evidence.	vbms.ui.veteran.enableTeraMemo
Auto-Population Button	Enables ability to toggle on / off the Auto-Population capability from the TERA Memorandum Form for question 1,2A,3,4,6.	AUTO_POPULATE_ENABLED
TERA Metric Gathering	Enables ability to toggle on / off the TERA Memorandum Form metric gathering to database.	METRICS_ENABLED
TERA Legacy	Enables ability to show the TERA Memo MFE in place of the legacy version of the page.	vbms.ui.mfeAdapter.teraMemo.enabled
TERA Auto-populate Metric Gathering	Enables ability to toggle on / off the TERA Memorandum Form metric gathering for auto-populate responses to database.	WRITE_AUTO_POP_TRACKING_FIELDS
TERA Memo 2A Auto-Population	Enables ability to toggle on / off the Auto-Population capability for TERA Memorandum form question 2A.	AUTO_POPULATE_2A_ENABLED
TERA Memo 6 Auto-Population	Enables ability to toggle on / off the Auto-Population capability for TERA Memorandum form question 6.	AUTO_POPULATE_6_ENABLED
Annotator Document Viewer	Enables ability to use annotator document viewer in VBMS.	OPEN_EVIDENCE_DOCS_IN_ANNOTATOR_VIEW
Return EDIPI	Returns EDIPI from Veteran API	GET_EDIPI_ENABLED
ILER Call	Enables ILER call from CE API	tera.memo.iler.enabled
TERA Auto-Populated Question Indicators	Defines which questions display the AI-generated hint text in the TERA Memorandum Form UI, indicating they may be auto-filled based on available evidence.	AUTO_POPULATED_QUESTIONS_IDS
TERA Multiple Highlights	Enables the ability to handle multiple highlights per supporting document in the TERA Memorandum Form instead of a single highlight per document in VBMS Claim Evidence API.	tera.memo.data.enabled
TERA Memo Data	Enables the ability to toggle on / off the TERA Memorandum Form auto-population capability and all related functionality in VBMS Claim Evidence API.	tera.memo.multipleHighlights.enabled

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Key Properties	Purpose	Location (Github)
TERA Import URL - MFE	The import URL for the TERA Memo MFE.	vbms.ui.mfeAdapter.teraMemo.import
CE_URL	The URL for Claim Evidence API.	https://github.com/department-of-veterans-affairs/bip-cpui-config
CLAIMS_URL	The URL for Claims API.	https://github.com/department-of-veterans-affairs/bip-cpui-config
VET_URL	The URL for Veteran API.	https://github.com/department-of-veterans-affairs/bip-cpui-config
DOCGEN_URL	The URL for DocGen Genstore API.	https://github.com/department-of-veterans-affairs/bip-cpui-config
VEFS_CE_URL	The URL for opening document links in a separate tab.	https://github.com/department-of-veterans-affairs/bip-cpui-config/blob/development/charts/bip-cpui-tera-memo-ui/dev/values-dev.yaml#L36

System Monitoring and Metrics

- The ML-MATS Platform provides system monitoring and metrics for TERA Memorandum in the below dashboard:
 - [Monitoring and Dashboards](#)

System Logging and Auditing

- Audits follow standard BIP Platform Auditing practices.

Security Strategy

- Security follows standard BIP Platform Security practices.
 - Compliance Scanning
 - Aligned with VBMS Resiliency Team regarding required security scanning standards on BIP Platform.
 - CodeQL
 - PrismaCloud
 - Rollback Procedures
 - In the case of a deployment related issue that is unable to be fixed within a given maintenance window, TERA Automation ART will revert to previous working versions and leverage feature flags accordingly.
 - Backup Strategy
 - Assets
 - All development code is stored in Github and version controlled.
 - All data is stored within the respective environment VBMS database tables that have their appropriate backup strategies in place.
 - In the case of issues with the TERA Memorandum Form
 - UI Issues will result in a feature flag of the entire form toggled off from VBMS with a communication from VA Leadership to end users on instructions to leverage manual PDF provided by VBA & Compensation Services.
 - Auto-Population issues will result in a feature flag toggled off for the Auto-Population Button along with communication to end users to continue use of the digital TERA Memorandum form on VBMS without auto-population functionality by VBA & Compensation Services.
 - In the case of BIP Platform infrastructure downtime, we will coordinate with both BIP Platform System Team & VBMS System Team to re-enable the TERA Memorandum form accordingly.
- TERA Memo Permissions
 - Permissions for accessing TERA Memorandum Form in VBMS have been granted based on users having the permissions to upload documents.
 - Users without proper permissions are unable to view the "TERA Memo" button in the actions dropdown of the Veteran profile page.
 - Users without proper permissions are unable to access the TERA Memorandum Form via direct URL.